Abedoulaye Dia

Philadelphia, PA | 445-318-7422 | abedoulaye.m.dia@gmail.com |LinkedIn | Github | Portfolio

EDUCATION

Temple University *Philadelphia, PA*
B.S. in Computer Science *Expected May 2029*
GPA: 3.3/4.0 | Relevant Coursework: Data Structures & Algorithms, Program Design & Abstraction, Mathematics.
Recipient of the Temple Promise Fellowship/Grant – Full-tuition institutional award granted to select Philadelphia graduates.

TECHNICAL SKILLS

Languages: Go, Python, Java, Typescript, JavaScript, SQL
Frameworks/Tools: React, Django, net/http, Git, Docker
Concepts: Data Structures, Algorithms, REST API, Object-Oriented Design

Work Experience

Customer Service Associate, Family Dollar — Philadelphia, PA *Sept 2025 – Jan 2026*

- Worked across all store functions — register, stocking, receiving, and floor maintenance — adapting to whatever needed.
- Handled cash, card, and return transactions for 100+ customers per shift with zero register discrepancies.
- Balanced part-time work with a full course load, including evening and morning classes.

PROJECTS

Diariata's Fabrics | *Go, net/http, React, PostgreSQL* Live Demo · Github

- Built a full-stack e-commerce platform for my mother's West African clothing business, enabling customers to browse products, manage carts, and complete secure payments through Stripe checkout.
- Engineered a JWT authentication system with access/refresh token rotation, transaction-based order processing with stock validation, and IP-based rate limiting to prevent brute-force attacks.
- Integrated Stripe PaymentIntents with webhook signature verification for payment confirmation, Resend for email verification, and deployed with input validation, structured error logging, and a responsive React frontend.

AI Coding Agent | Python, OpenAI SDK, Subprocess, argparse · Github

- Built an autonomous debugging agent that iteratively reads, writes, and executes code to fix issues without human intervention, using OpenAI's function-calling API and a multi-step feedback loop of up to 20 iterations.
- Engineered a sandboxed file system layer using os.path.commonpath validation to confine all LLM file operations to a configurable working directory, preventing unauthorized access outside the project scope.
- Designed 4 JSON Schema-declared tool functions (list, read, write, execute) enabling the LLM to dynamically select and invoke the appropriate action, and added a --verbose CLI flag to surface token usage, function call arguments, and raw tool outputs during execution.

CoinSight - Cryptocurrency Dashboard | *Typescript, React, CoinGecko API, Echarts* Live Link · Github

- Built a real-time cryptocurrency dashboard displaying live market data for 100+ coins with interactive charts and individual coin detail pages, as a personal project.
- Applied the CoinGecko API with ECharts 7-day line charts and interactive tooltips to deliver responsive, drill-down market visualization across every listed coin.
- Engineered an investment calculator computing historical returns (e.g., "If you invested $100 365 days ago you'd have __") and AI-generated per-coin market summaries, surfacing insights beyond raw price data.